

SEC. 3 PROPOSED APPROACH

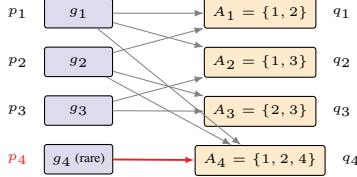


Fig. 1. The mask E . Batch compositions carry the mass q they occur with; groups must receive the mass p the objective assigns them. Group 4 is reachable only through A_4 , so exact alignment requires $p_4 \leq q_4$: the Hall condition of Thm. 1 read on the singleton $I = \{4\}$. Shrinking the batch deletes A_4 from the support and makes p_4 unreachable.

1. PROPOSED APPROACH: ALIGNMENT BY MASKED OPTIMAL TRANSPORT

Weights as a coupling. Let $Y[i, j] \geq 0$ be the weight assigned to group i when the active set is A_j , with $Y[i, j] = 0$ for $i \notin A_j$, and let $T[i, j] \triangleq q_j Y[i, j]$ be the joint mass carried from composition j to group i . Requiring that the weights be a probability vector on each realized composition, and that each group receive in expectation the mass p assigns it, is the *masked optimal transport* (MOT) feasibility problem of [6],

$$T\mathbf{1}_M = p, \quad \mathbf{1}_N^\top T = q, \quad T \geq 0, \quad \text{supp}(T) \subseteq E, \quad (1)$$

with mask $E = \{(i, j) : i \in A_j\}$ encoding that a group absent from the batch cannot be weighted. Unlike classical transport [25, 20] there is no cost to minimize: any coupling respecting (1) serves, and (1) is the Kantorovich problem with the indicator cost $C[i, j] = \infty$ off E . Fig. 1 is the picture.

When alignment is possible. Existence of a coupling is a supply–demand question on the bipartite graph E , and is settled by flow duality.

Theorem 1 (Feasibility, [6, Thm. 1]). *For $I \subseteq [N]$ let $\mathcal{N}(I) = \{j : \exists i \in I, (i, j) \in E\}$ and $\mathcal{S}(I) = \{j : A_j \subseteq I\}$. Problem (1) is feasible if and only if*

$$\sum_{j \in \mathcal{S}(I)} q_j \leq \sum_{i \in I} p_i \leq \sum_{j \in \mathcal{N}(I)} q_j \quad \forall I \subseteq [N]. \quad (2)$$

Proofs are in the appendix. Read on singletons, (2) becomes a statement one can act on before training starts.

Corollary 2 (Coverage law and severity). *Feasibility of (1) requires $\mathbb{P}[A^t = \{i\}] \leq p_i \leq r_i$ for every i , with $r_i = \mathbb{P}[i \in A^t]$ the observation rate of Sec. 2. Writing*

$$\nu \triangleq \sum_{i : p_i > r_i} p_i \quad (3)$$

Algorithm 1 GAVOT: group-fair SGD by masked transport

Require: target p , composition law $(q, \{A_j\}, E)$, step η , horizon T , CVaR parameters (α, γ) with $\gamma = 1$ for the risk-neutral method

- 1: $(T, Y) \leftarrow$ Sinkhorn scaling for (p, q, E) ▷ offline, once
- 2: $\theta^0 \in \mathcal{C}$, $t^0 \in \mathbb{R}$
- 3: **for** $t = 1, \dots, T$ **do**
- 4: draw batch B^t ; set A^t , let $j(t)$ be s.t. $A_{j(t)} = A^t$
- 5: $g_i^t \leftarrow$ stochastic subgradient of f_i from $B^t \cap$ group i
- 6: $w_i^t \leftarrow Y[i, j(t)](\gamma + \frac{1-\gamma}{\alpha} \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$
- 7: $\theta^t \leftarrow \Pi_{\mathcal{C}}(\theta^{t-1} - \eta \sum_{i \in A^t} w_i^t g_i^t)$
- 8: $t^t \leftarrow t^{t-1} - \eta(1 - \gamma)(1 - \frac{1}{\alpha} \sum_{i \in A^t} Y[i, j(t)] \mathbf{1}\{\hat{f}_i^t > t^{t-1}\})$
- 9: **end for**

Ensure: $\bar{\theta}_T = \frac{1}{T} \sum_{t=1}^T \theta^t$

for the infeasible mass, $\nu = 0$ iff the upper constraints hold. Under i.i.d. sampling of b examples the upper constraint reads $b \geq \log(1 - p_i)/\log(1 - \pi_i)$; under K -of- N uniform group sampling it reads $p_i \leq K/N$.

Severity. Importance may not exceed observation rate: a rare group is reachable only if the batch is large enough to contain it often enough. The infeasible mass ν is the axis we sweep: at $\nu = 0$ exact alignment is available, and as ν grows more of p sits outside the set of row marginals the mask can reach. A *feasible* and an *infeasible* regime are two points on this continuum, and Sec. 5 sweeps it instead.

Computing the plan. Given feasibility, a coupling is obtained by iterative proportional fitting (Sinkhorn scaling) on the masked support [18, 19, 20]: starting from any $T^{(0)} > 0$ on E with $\mathbf{1}^\top T^{(0)} = q$, alternate

$$T \leftarrow \text{diag}(p \oslash T\mathbf{1}) T, \quad T \leftarrow T \text{diag}(q \oslash T^\top \mathbf{1}) \quad (4)$$

until $\|T\mathbf{1} - p\|_1 \leq \varepsilon$. Each sweep costs $O(|E|)$ with $|E| \leq NM$ and M the number of *realized* compositions rather than 2^N ; the plan is computed once, offline, and the training loop adds one lookup and one multiply per group per step. Under (2) the iteration converges to a solution of (1) [18, 22], being block-coordinate ascent on the dual of the entropic problem $\min\{\sum_E T \log T : T\mathbf{1} = p, \mathbf{1}^\top T = q\}$. Since $\mathbf{1}_N^\top T = q$ and $Y = T \text{diag}(q)^{-1}$, every column $Y[\cdot, j]$ is a probability vector on A_j , the property each bound below uses. Algorithm 1 is the resulting method; $\gamma = 1$ is the risk-neutral rule analyzed here and $\gamma < 1$ the risk-averse layer of Sec. 4.

Analysis. Everything rests on one identity, [6, Lem. 1] moved from model averaging to gradient averaging. Let $g_i(\theta)$ be the stochastic subgradient of f_i computed from $B^t \cap$ group i and $\hat{g}(\theta) = \sum_{i \in A^t} Y[i, j(t)] g_i(\theta)$ the transport-weighted batch subgradient of Algorithm 1. Transport removes bias; it is not a variance reduction device.

Lemma 3 (Unbiasedness). *Let T satisfy (1). Then $\mathbb{E}[\hat{g}(\theta) \mid \theta] \in \partial F_p(\theta)$ for every $\theta \in \mathcal{C}$.*

Proof. Conditionally on $A^t = A_j$, $\mathbb{E}[g_i(\theta)] \in \partial f_i(\theta)$ for $i \in A_j$. Averaging over $A^t \sim q$, $\mathbb{E}[\hat{g}] = \sum_j q_j \sum_{i \in A_j} Y[i, j] \partial f_i(\theta) = \sum_i (\sum_j T[i, j]) \partial f_i(\theta) = \sum_i p_i \partial f_i(\theta)$, the row-marginal constraint being used in the last step. $\square$

Lemma 4 (Second moment, free of coverage). $\mathbb{E}\|\hat{g}(\theta)\|^2 \leq G^2$ for every $\theta \in \mathcal{C}$, with no dependence on $|A^t|$ or on N .

Proof. $Y[\cdot, j]$ is a probability vector on A_j , so $\|\sum_i Y[i, j] g_i\|^2 \leq \sum_i Y[i, j] \|g_i\|^2$ by convexity of $\|\cdot\|^2$; take expectations and use $\sum_j T[i, j] = p_i$: $\mathbb{E}\|\hat{g}\|^2 \leq \sum_i p_i \mathbb{E}\|g_i\|^2 \leq G^2$. $\square$

Theorem 5 (Rate). *Under Assumption 1, if (1) is feasible then Algorithm 1 with $\gamma = 1$ and $\eta = D/(G\sqrt{T})$ satisfies*

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) \leq \frac{DG}{\sqrt{T}}. \quad (5)$$

The constant in (5) is that of full-participation SGD and is independent of how many groups appear in a batch: a composition covering two groups yields the same bound as one covering all N , though with larger realized variance. The proof uses only the two lemmas and never refers to $|A^t|$, which is why the rate of [6] transfers verbatim from aggregation of models to aggregation of gradients.

Outside the feasible region. When (2) fails, (1) has no solution and the iteration (4) does not diverge: its iterates accumulate at the coupling whose row marginal $\hat{p} \triangleq T\mathbf{1}$ is the KL-closest to p over the reachable polytope $\mathcal{R}(q, E) = \{T\mathbf{1} : T \geq 0, \mathbf{1}^\top T = q, \text{supp}(T) \subseteq E\}$ [22, 21, 20]. The column marginal is met exactly, so Lemmas 3 and 4 hold with p replaced by $\hat{p}$: Algorithm 1 is then unbiased SGD on the surrogate $F_{\hat{p}}$, the closest objective the sampler can reach, and the loss on (1) is a bias of exactly the shape of Proposition 2. On singletons $\hat{p}_i \leq r_i$, so the groups with $p_i > r_i$ are truncated at their observation rate and their missing mass, ν up to renormalization, is redistributed.

Theorem 6 (Infeasible regime). *Let $\hat{p}$ be the row marginal of the Sinkhorn limit, $\theta^\dagger \in \arg \min_{\mathcal{C}} F_{\hat{p}}$, and write $\Phi(w) \triangleq F_p(\arg \min_{\mathcal{C}} F_w)$ for the p -risk of the minimizer of the w -weighted risk. Then*

$$\mathbb{E}[F_p(\bar{\theta}_T)] - F_p(\theta^*) = \underbrace{\Phi(\hat{p}) - \Phi(p)}_{\text{alignment floor}} + \underbrace{\varepsilon_T(\hat{p})}_{\text{optimization residual}}, \quad (6)$$

where $\varepsilon_T(\hat{p}) \triangleq \mathbb{E}F_p(\bar{\theta}_T) - F_p(\theta^\dagger)$ satisfies $\mathbb{E}F_{\hat{p}}(\bar{\theta}_T) - F_{\hat{p}}(\theta^\dagger) \leq DG/\sqrt{T}$ and $\varepsilon_T(\hat{p}) \rightarrow 0$ whenever $\bar{\theta}_T \rightarrow \theta^\dagger$. Group-blind averaging obeys (6) with $\tilde{p}$ in place of $\hat{p}$. Writing $\Delta_{\text{floor}} \triangleq \Phi(\tilde{p}) - \Phi(\hat{p})$ for the alignment gain, transport improves on group-blind averaging exactly when $\Delta_{\text{floor}} > \varepsilon_T(\hat{p}) - \varepsilon_T(\tilde{p})$, and every floor is bounded by $\Phi(w) - \Phi(p) \leq 2M\|p - w\|_1$.

Theorem 6 replaces the usual bias *bound* [6] by an exact *decomposition*, both terms of which are available before training: $\hat{p}$ and $\tilde{p}$ from (p, q, E) , and Φ exactly on instances with a closed-form or convex frontier. The gain is fixed by the instance and decides the sign of the comparison; the residual is paid by the estimator and, as Sec. 5 shows, is a stepsize artifact. The ℓ_1 bound is the weaker object and we report both, because it turns out too loose to discriminate.